3.4.2 Solenoid

A current carrying wire is made to form a loop and a number of loops are wound in the same direction to form a coil. More magnetic fields will add to make the flux lines through the coil stronger and dense.

A helically wound coil that is made to produce a strong magnetic field is called a solenoid.

The flux lines in a solenoid act in the same way as in a magnet. They leave the north pole and go around to the south pole.

The directions of the magnetic field in a solenoid is known by the following rules.

End Rule

Look at the end of the solenoid of the electromagnet. If the current in the coil is clock wise the end is Southpole . If the current in the coil is counter-clockwise the end is North pole.

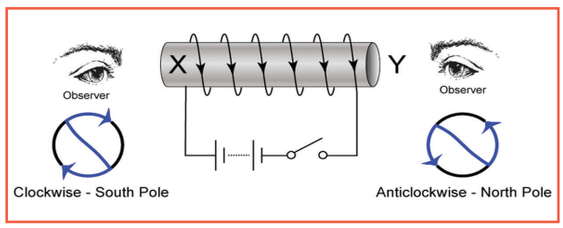


Fig 3.10 *End Rule*

Helix Rule

Hold the right hand palm over the solenoid in such a way the fingers point in the direction of current in the solenoid conductors. Then the thumb indicates the direction of magnetic field (North) of the solenoid.

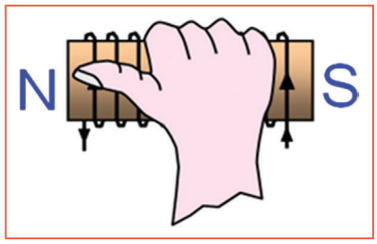


Fig 3.11 *Helix Rule*

Uses of Solenoid

- Used for Circuit Breaking.
- Voltage Regulating Device.
- Automatic Motor Starter.
- Contactor, Elevator, Crane.

Toroid

A helix bent into a circular form is known as Toroid (i.e coiled coil)



Fig 3.12 *Toroid*

3.5 ELECTROMAGNETIC INDUCTION

Electricity induced by the magnetic field is known as Electro Magnetic Induction.

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